FIITJEE

JEE (Advanced)-2018

ALL INDIA TEST SERIES

FULL TEST – X

Paper 1

Time Allotted: 3 Hours **Maximum Marks: 183**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part is further divided into Two sections: **Section-A & Section-C**
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

1. ***Section-A (01– 07, 19 – 25, 37 - 43)*** *contains 21 multiple choice questions which have* ***one or more than one correct*** *answer. Each question carries* ***+4 marks*** *for correct answer and* ***–2 marks*** *for wrong answer*
   Partial Marks **+1** for each correct option provided no incorrect options is selected.

   Section-A (08 – 13, 26 – 31, 44 - 49) *contains 18 questions. Each of 2 Tables with 3 Columns and 4 Rows has three questions. Column 1 will be with 4 rows designated (I), (II), (III) and (IV). Column 2 will be with 4 rows designated (i), (ii), (iii) and (iv). Column 3 will be with 4 rows designated (P), (Q), (R) and (S).*
   Each question has ***only one correct*** *answer and carries* ***+3 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

2. ***Section-C (14 – 18, 32 – 36, 50 - 54)*** *contains 15 Numerical based questions with answer as numerical value from* ***0 to 9*** *and each question carries* ***+3 marks*** *for correct answer. There is no negative marking.*

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

Physics

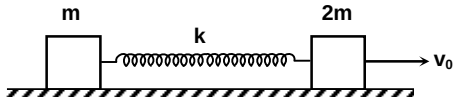
PART – I

SECTION – A

Straight Objective Type

This section contains **7 multiple choice questions.** Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which only **ONE OR MORE THAN ONE** is/are correct

1. Two blocks of mass m and 2m are connected by a spring of spring constant k. The system is placed on a horizontal frictionless surface. Now at t =0, the block 2m is projected horizontally with velocity v_0 towards right. Then



(A) Minimum time when the elongation in spring is maximum is $t = \frac{\pi}{2}\sqrt{\frac{2m}{3k}}$.

(B) The minimum speed of block of mass 2m is $\frac{v_0}{3}$.

(C) The maximum elongation in the spring is $\sqrt{2mv_0^2/3k}$.

(D) The amplitude of oscillation of block of mass 2m with respect the centre of mass frame of the system is $\sqrt{\frac{2mv_0^2}{27k}}$.

2. Gravitational constant G, Bulk modulus B and coefficient of viscosity η are used to form a unit of mass M and length L. Then the correct option(s) is/are

(A) $M \propto \eta^{3/2}$ (B) $M \propto G^{1/2}$

(C) $L \propto \eta^2$ (D) $L \propto B^{-3/2}$

Space for Rough work

3. An object of mass m moving with velocity v, explodes suddenly in two parts of mass ratio 1 : 2. If the direction of motion of two parts makes an angle 60° with each other after explosion and de Broglie wavelength of the lighter part is half of the de-Broglie wavelength of the heavier part, then

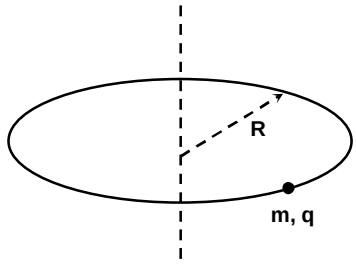
(A) Speed of the smaller part after explosion is $\frac{6v}{\sqrt{7}}$.

(B) Speed of the heavier part after explosion is $\frac{3v}{2\sqrt{7}}$.

(C) Increase in kinetic energy of the system due to explosion is $\frac{4}{7}mv^2$

(D) Ratio of kinetic energy of lighter part to that of heavier part is 4 : 1

4. A small bead of mass m and charge q is threaded on a thin horizontal ring of radius R made of insulating material. The bead can move on the circular track without friction and is initially at rest. A magnetic field that is cylindrically symmetric about the axis of the ring is present and directed along the axis. The magnitude of magnetic field at a distance r from the axis and at time t is given as $B = \frac{K}{r}t$, where K is a constant.



Then

(A) Velocity of the bead at time t is $v = \frac{Kqt}{2m}$

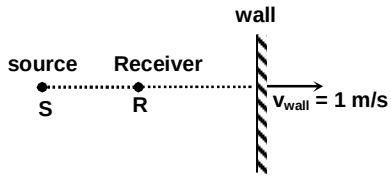
(B) Velocity of the bead at time t is $v = \frac{Kqt}{m}$

(C) Normal force acting on the bead at time t is N = 0

(D) Normal force acting on the bead at time t is $N = \frac{K^2q^2t^2}{mR}$

Space for Rough work

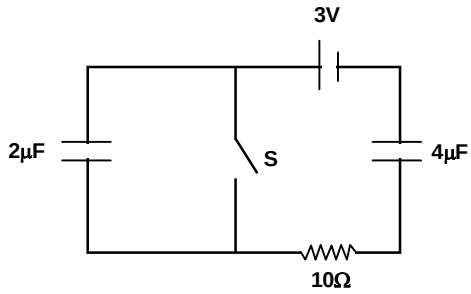
5. A source of sound of frequency f = 1000 Hz and a receiver are located on the same normal to a wall. The wall recedes the receiver with a velocity 1 m/s. Velocity of sound in air is $v = \frac{1000}{3}$ m/s.



(A) The wavelength of sound changes by 0.6% after reflection from the wall.
(B) The wavelength of sound changes by 0.3% after reflection from the wall..
(C) The frequency of echo heard by the receiver is 994 Hz.
(D) The beat frequency detected by the receiver is 6 Hz.

6. A stationary Pb^{206} nucleus emits an α-alpha particle with kinetic energy 5.77 MeV. Then
(A) The recoil speed of the daughter nucleus is approximately equal to 3.4×10^5 m / s .
(B) The total energy released in the decay is approximately equal to 5.88 MeV.
(C) The recoil speed of the daughter nucleus is approximately equal to 6.8×10^5 m/s.
(D) The total energy released is approximately equal to 6.22 MeV.

7. The circuit shown in figure consists of two capacitors of capacitance 2μF and 4μF, a resistor of resistance R = 10 Ω and an ideal battery of emf 3V. Initially the switch S was open for a long time and the capacitors were fully charged. Then after the switch S is closed,
(A) Heat produced in the circuit after the switch is closed is 12 μJ.
(B) Energy supplied by the battery after the switch is closed is 24 μJ



(C) Change in energy stored in the capacitor of capacitance 4 μF after the switch is closed is 16 μJ
(D) Charge flown through the battery after the switch is closed is 12 μC

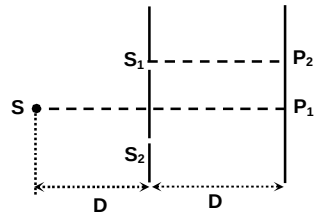
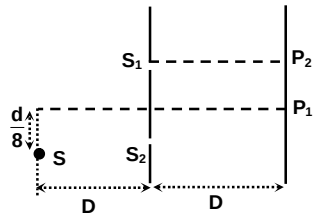
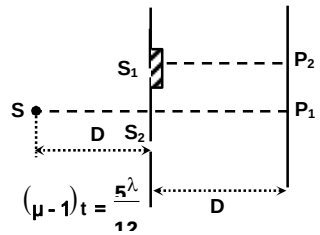
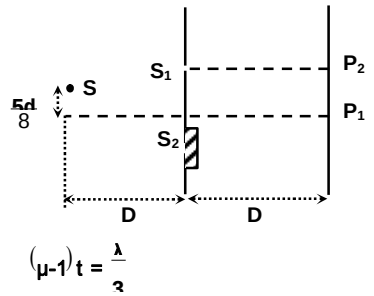
Space for Rough work

Matrix Match Types

Each of 2 tables with 3 Columns and 4 Rows has three questions. Column-1 will be with 4 rows designated (I), (II), (III) and (IV). Column-2 will be with 4 rows designated (i), (ii), (iii) and (iv). Column-3 will be with 4 rows designated (P), (Q), (R) and (S). Each question has four options with only **ONE** correct

Answer questions 8, 9 and 10 by appropriately matching the information given in the three columns of the following table.

Column-1 shows four situations of standard Young's double slit arrangement with the screen placed at a distance D = 1.5 m from the slits S_1 and S_2. The distance between the slits is d = 1 mm and wavelength (in air) of light source S used in the experiment is λ = 500 nm. The maximum fringe intensity on the screen is I_m. In column-2 $\delta(P)$ denote the path difference at point P and in column-3 I(P) denote the intensity at point P. A transparent sheet of thickness t and refractive index μ is placed in front of one of the slits in third and fourth cases of column-1 as shown in the figure.

Column 1		Column 2		Column 3	
(I)		(i)	$\delta(P_2)=\frac{2\lambda}{3}$	(P)	$I(P_2)=\frac{I_m}{2}$
(II)		(ii)	$\delta(P_1)=\frac{\lambda}{6}$	(Q)	$I(P_2)=I_m\cos^2\frac{\pi}{12}$
(III)		(iii)	$\delta(P_2)=\frac{\lambda}{4}$	(R)	$I(P_2)=\frac{I_m}{4}$
(IV)		(iv)	$\delta(P_2)=\frac{11\lambda}{6}$	(S)	$I(P_1)=\frac{3I_m}{4}$

Space for Rough work

8. Which of the following option is the only correct representation of a situation in which $\delta(P_1) = \frac{5\lambda}{12}$

(A) (I) (i) (Q) (B) (II)(ii) (S)
(C) (III) (iii) (P) (D) (IV) (iv) (R)

9. Which of the following option is the only correct representation of a situation in which $I(P_2) = 0$.

(A) (I) (i) (S) (B) (II)(ii) (S)
(C) (III) (iii) (Q) (D) (IV) (iv) (R)

10. Which of the following option is the only correct representation of a situation in which $\delta(P_1) = 0$.

(A) (I) (i) (S) (B) (II)(ii) (R)
(C) (I) (i) (R) (D) (IV) (iv) (Q)

Space for Rough work

Answer questions 11, 12 and 13 by appropriately matching the information given in the three columns of the following table.

Column 1		Column 2		Column 3	
(I)	An ideal diatomic gas is compressed such that its pressure $P \propto T^{7/2}$ where T is temperature of the gas	(i)	Heat is rejected by the gas	(P)	Bulk modulus of the gas for the process is $B = \frac{3P}{2}$.
(II)	An ideal monatomic gas is compressed such that $P \propto V^{-3/2}$ where V is volume of the gas	(ii)	Heat supplied to the gas is zero	(Q)	Coefficient of volume expansion of the gas = $\frac{1}{2T}$.
(III)	An ideal diatomic gas expanded such that its internal energy $U \propto \frac{1}{\rho}$ where ρ is density of the gas	(iii)	Pressure of gas remains constant	(R)	The Bulk modulus of the gas for the process is B = 0.
(IV)	An ideal monatomic gas expands such that its pressure $P \propto V$ where V is the volume of the gas	(iv)	Its temperature versus pressure graph is a parabola	(S)	Compressibility of the gas for the process $= \frac{5}{7P}$.

11. Which of the following option is the only correct representation of the process in which $\Delta U = -\int PdV$?
(A) (II) (i) (S) (B) (III) (ii) (R)
(C) (I) (ii) (P) (D) (I) (ii) (S)

12. Which of the following option is the only correct representation in which molar heat capacity of the gas is negative?
(A) (II) (iii) (P) (B) (II) (i) (P)
(C) (IV) (iii) (Q) (D) (I) (ii) (R)

13. Which of the following options is the correct combination in which $\Delta U = \Delta Q - P\Delta V$?
(A) (IV) (iii) (Q) (B) (I) (ii) (Q)
(C) (III) (iii) (R) (D) (I) (i) (Q)

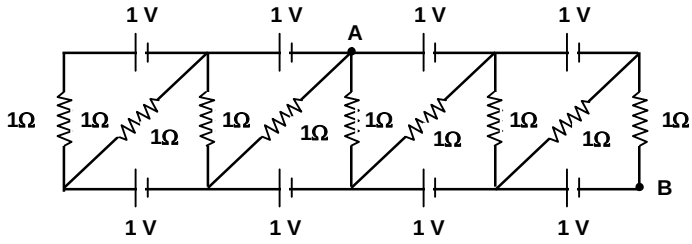
Space for Rough work

SECTION – C

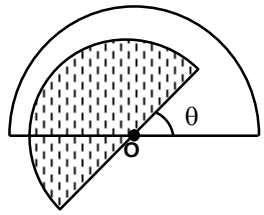
(One Integer Value Correct Type)

This section contains **5 questions**. Each question, when worked out will result in **one integer** from 0 to 9 (both inclusive).

14. If the potential difference $V_A - V_B$ for the given circuit equals $\frac{22}{n}$ volts find the value of n. All the cells are ideal, and emf of each cell is 1 Volt.

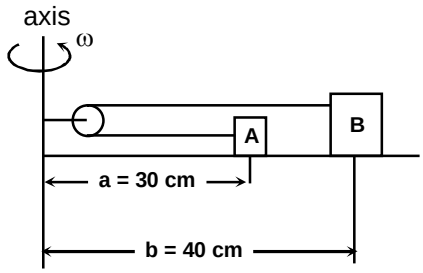


15. A capacitor consists of two fixed semicircular plates of radius R and a movable slab of thickness d and dielectric constant k placed between then. The slab can freely rotate about the axis O and practically fills the entire gap between the fixed plate. A constant voltage V is maintained between the plates. If the torque acting on the dielectric slab when the slab is displaced to an angle θ as shown in the figure, due to electrostatic force is $\tau = \frac{(k-1)\varepsilon_0 R^2 V^2}{nd}$ find the value of n.

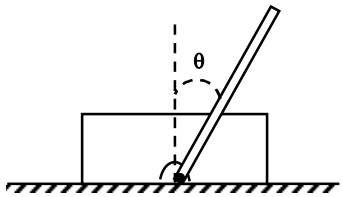


Space for Rough work

16. Block A has a mass $m_A = 15\,kg$ and block B has mass of $m_B = 45 kg$. They are placed on a rotating horizontal surface and are connected to a massless string passing through a frictionless pulley as shown in the figure. The horizontal surface is rotating about a vertical axis with uniform angular velocity ω as shown in the figure. The coefficient of friction between the blocks and the surface is μ = 0.25. If the value of angular speed of the surface at which the blocks will start sliding on the surface is $\omega = \frac{10}{n}$ rad/sec, find the value of n. (given g = 10 m/s^2)



17. A rod of mass m and length $L = \frac{25}{6}m$ is hinged in a plank of same mass m. The plank is kept on a smooth horizontal surface and rod makes angle θ with the vertical. The system is released from rest from θ = 0°. The velocity (in m/s) of the plank when the angle θ = 180° is (Take g = 10 m/s^2).



18. The nuclei of a radioactive element A decay into nuclei of radioactive element B with decay constant λ. The nuclei of B further decay into stable nuclei of an element with decay constant 2λ. At t = 0 the number of nuclei of A was $N_0 = 2\times10^{21}$ and that of B was zero, if the number of nuclei of B when its activity is maximum is $p\times10^{20}$ then find the value of p

Space for Rough work

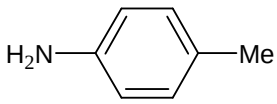
Chemistry PART – II

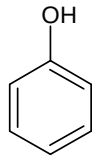
SECTION – A

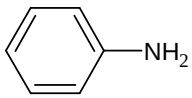
Straight Objective Type

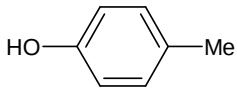
This section contains **7 multiple choice questions.** Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which only **ONE OR MORE THAN ONE** is/are correct

19. p-Toluidine reacts with benzenediazonium chloride to form a compound, which on boiling with H_2SO_4, gives product/s:

(A) 

(B) 

(C) 

(D) 

20. Which of the following cation/s do/does not give coloured bead in borax bead test?
(A) Cu^{+2} (B) Co^{+3}
(C) Cd^{+2} (D) Al^{+3}

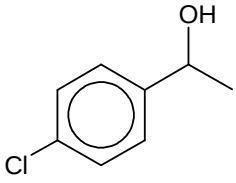
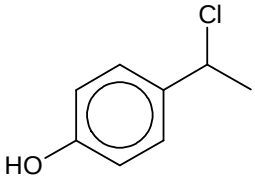
21. Which of the following are true for Bakelite?
(A) Cross linkage polymer. (B) Condensation polymer.
(C) Thermosetting polymer. (D) Addition polymer.

22. Which of the following species is/are planar and non-polar with lone pair/s, of electrons on the central atom?
(A) ClF_3 (B) XeF_5^-
(C) PCl_5 (D) BrF_5

Space for Rough work

23. Which of the following statement/s, is/are incorrect?
(A) Complex $Na_2[Fe(CN)_5NO].2H_2O$ is formed during brown ring test for nitrate.
(B) The oxidation number of Fe in the above given complex is +3.
(C) Complex $[Fe(H_2O)_5NO]^{+2}$ is brown in colour.
(D) Oxidation number of Fe in sodium nitroprusside complex is +1.

24. Which of the following statement is/are not correct regarding $CuSO_4$?
(A) It reacts with KI to give iodine.
(B) It reacts with KCl to give Cu_2Cl_2.
(C) It reacts with NaOH and glucose to give Cu_2O.
(D) It gives CuO on strong heating in air.

25. Which of the following is/are correct option(s):
(A)  and  can be differented by victor Mayer's test.
(B) Monomer of orlon is acrylonitrile.
(C) Aspartane is a dipeptide of L-aspartic acid and L-phenylalanine.
(D) D-Glucose, D-Fructose and L-Mannose give same osazone with Ph – NH – NH_2.

Space for Rough work

Matrix Match Types

Each of 2 tables with 3 Columns and 4 Rows has three questions. Column-1 will be with 4 rows designated (I), (II), (III) and (IV). Column-2 will be with 4 rows designated (i), (ii), (iii) and (iv). Column-3 will be with 4 rows designated (P), (Q), (R) and (S). Each question has four options with only **ONE** correct

Answer questions 26, 27 and 28 by appropriately matching the information given in the three columns of the following table.

Column 1		Column 2		Column 3	
(I)	Smelting	(i)	Copper glance	(P)	Cu_2S
(II)	Self reduction	(ii)	Malachite	(Q)	Fe_2O_3
(III)	Electrolytic reduction	(iii)	Haematite	(R)	$Al_2O_3.2H_2O$
(IV)	Hydrometallurgy	(iv)	Bauxite	(S)	$CuCO_3.Cu(OH)_2$

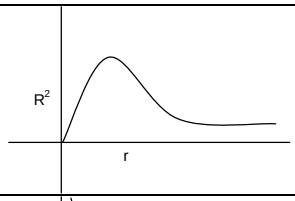
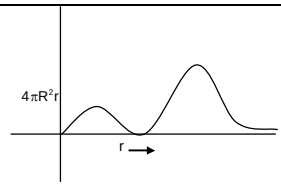
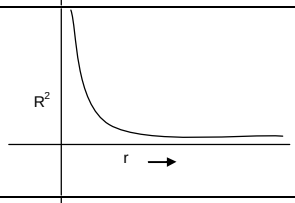
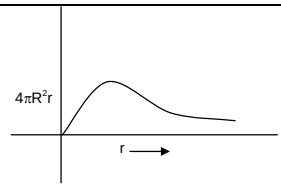
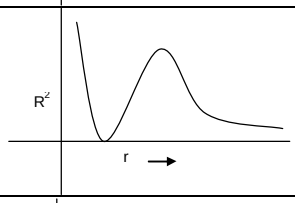
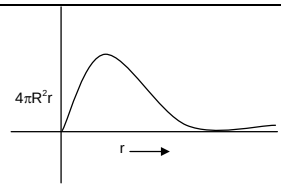
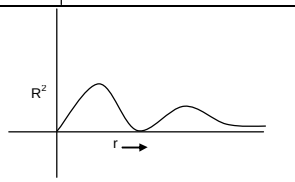
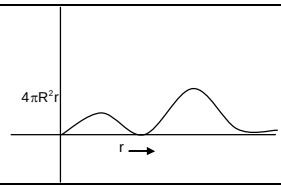
26. Which of the following combination is CORRECT for Copper glance?
(A) (I) (i) (S) (B) (II)(i) (P)
(C) (III) (i) (S) (D) (IV) (i) (P)

27. Which of the following combination is CORRECT for malachite?
(A) (I) (ii) (R) (B) (IV) (ii) (P)
(C) (IV) (ii) (S) (D) (III) (ii) (S)

28. Which of the following combination is CORRECT for Hematite?
(A) (II) (iii) (Q) (B) (III) (iii) (R)
(C) (III) (iii) (Q) (D) (I) (iii) (Q)

Space for Rough work

Answer questions 29, 30 and 31 by appropriately matching the information given in the three columns of the following table.

Column 1		Column 2		Column 3	
(I)	1s orbital	(i)		(P)	
(II)	2s orbital	(ii)		(Q)	
(III)	2p orbital	(iii)		(R)	
(IV)	3p orbital	(iv)		(S)	

29. Which of the following combination is CORRECT for 2s-orbital?
(A) (II) (ii) (R) (B) (II)(iii) (R)
(C) (II) (iii) (P) (D) (II) (iv) (S)

30. Which of the following combination is CORRECT for 2p-orbital?
(A) (III) (iii) (P) (B) (III) (iv) (Q)
(C) (III) (i) (S) (D) (III) (i) (Q)

31. Which of the following combination is CORRECT for 3p-orbital?
(A) (IV) (iii) (S) (B) (IV) (iv) (S)
(C) (IV) (ii) (Q) (D) (IV) (iii) (Q)

Space for Rough work

SECTION – C

(One Integer Value Correct Type)

This section contains **5 questions**. Each question, when worked out will result in **one integer** from 0 to 9 (both inclusive).

32. Out of NiS, CoS, MnS, ZnS
A = Number of black precipitates
B = Number of sulphides which is/are soluble in HCl.
C = Number of sulphides which is/are soluble in CH_3COOH
D = Number of sulphides which is/are soluble in conc. HNO_3
What will be the value of A + B + C + D?

33. $PhCO_2H + NaHCO_3 \longrightarrow A(gas)$
$PhOH + NaNH_2 \longrightarrow B(gas)$
$CH_3OH + NaH \longrightarrow C(gas)$
$CH_3SH + EtLi \longrightarrow D(gas)$
$PhSO_3H + Na \longrightarrow E(gas)$
If the molecular weight of A, B, C, D, E is x, y, z, w, v. then find the value of (y + z + w + v) – x.

34. How many among the following species contain P – P bond(s):
(i) P (red) (ii) $H_4P_2O_5$ (iii) $H_4P_2O_7$ (iv) $\left(PO_3^-\right)_3$ (v) $\left(PO_3^-\right)_2$ (vi) P_4O_{10}
(vii) P_4S_3 (viii) P_4O_6 (ix) P_4

35. Identify the number of correct statements:
(i) All adiabatic process must be fast.
(ii) All reversible process must be slow.
(iii) For adiabatic process, dq = 0.
(iv) If q = 0, process must be adiabatic.
(v) Absolute value of internal energy can not be determined.
(vi) H = E + PV is applicable only for ideal gases.
(vii) $\Delta H_f^o\left(D_2(g)\right) = 0$, at 298 K.
(viii) $\Delta G_f^o\left(H^+aq\right) = 0$, at 298 K.
(ix) Expansion of an ideal gas not always represent an increase in entropy of system.

36. How many isomeric dienes with a six membered ring are possible of the compound with the molecular formula C_7H_{10}, while two consecutive double bonds are not possible in six membered ring.

Space for Rough work

Mathematics PART – III

SECTION – A

Straight Objective Type

This section contains **7 multiple choice questions.** Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which only **ONE OR MORE THAN ONE** is/are correct

37. The integral $\int_{\pi/4}^{\pi/2} (2\cot x)^{2017}\, dx$ is equal to

(A) $2\int_0^2 \frac{t^{2017}}{t^2+4} dt$

(B) $2\int_2^{2\sqrt{2}} \frac{(t^2-4)^{1008}}{t} dt$

(C) $\frac{2^{2012}}{63} - 4\int_4^8 \frac{(t-4)^{1007}}{t} dt$

(D) $2\int_0^{\ell n(1+\sqrt{2})} \left(\frac{e^t - e^{-t}}{e^t + e^{-t}}\right)^{2017} dt$

38. Sum of series $\frac{1}{\sqrt[3]{1}+\sqrt[3]{2}+\sqrt[3]{4}} + \frac{1}{\sqrt[3]{4}+\sqrt[3]{6}+\sqrt[3]{9}} + \frac{1}{\sqrt[3]{9}+\sqrt[3]{12}+\sqrt[3]{16}} + ...$ to 2196 terms is

(A) a rational number

(B) an irrational number

(C) less than 13

(D) $\sqrt[3]{2196} - 1$

39. Let $A = \tan^{-1}(\cot 65^\circ + 2\tan 40^\circ)$; $B = \cos^{-1}\left(\sqrt{3}\sin 80^\circ - 2\sin 50^\circ\right)$ then

(A) $\cos(2B - A) > 0$

(B) $\cot(A + B) > -1$

(C) $\sin(2A + 3B) > 0$

(D) $\tan(2A - B) > 1$

Space for rough work

40. Let f be a twice differentiable function on $(0, \infty)$ such that $f(x) > 0$, $f'(x) > 0$ and $f''(x) > 0 \ \forall \ x > 0$. If $\lim\limits_{x\to\infty} \dfrac{f'(x)f''(x)}{(f''(x))^2} = \dfrac{1}{3}$ then

(A) $\lim\limits_{x\to\infty} f(x) = \infty$

(B) $\lim\limits_{x\to\infty} \dfrac{f'(x)}{xf''(x)} = \dfrac{2}{3}$

(C) $\lim\limits_{x\to\infty} \dfrac{xf'(x)}{f(x)} = \dfrac{3}{2}$

(D) $\lim\limits_{x\to\infty} \dfrac{f(x)f''(x)}{(f'(x))^2} = 1$

41. The tangent to the circle $x^2 + y^2 = 4$ at a point P intersect the parabola $y^2 = 4x$ at points Q and R. Tangents to the parabola at Q and R intersect at S. If Q lies in the first quadrant such that PQ = 1 unit, then

(A) PR = 8 units

(B) $SR = \dfrac{35}{4}$ units

(C) $SQ = \dfrac{7}{2}$ units

(D) area of $\Delta SQR = \dfrac{343}{16}$ sq. units

42. Let $\vec{a}$, $\vec{b}$ and $\vec{c}$ be three non-coplanar vectors and $\vec{d}$ be a non-zero vector perpendicular to $\vec{a}+\vec{b}+\vec{c}$. Now $\vec{d} = (\vec{a}\times\vec{b})\sin x + (\vec{b}\times\vec{c})\cos y + 2(\vec{c}\times\vec{a})$, then

(A) $\dfrac{\vec{d}\cdot(\vec{a}+\vec{c})}{[\vec{a}\ \vec{b}\ \vec{c}]} = 2$

(B) $\dfrac{\vec{d}\cdot(\vec{a}+\vec{c})}{[\vec{a}\ \vec{b}\ \vec{c}]} = -2$

(C) minimum value of $x^2 + y^2 = \dfrac{\pi^2}{4}$

(D) minimum value of $x^2 + y^2 = \dfrac{5\pi^2}{4}$

43. If the two curves $y = ax^3 - 6ax^2 + (12a + 12)x - (8a + 16)$ where $a \in R$ and $y = x^3$ touches each other at some point, then possible value of 'a' are

(A) 0

(B) 1

(C) $\sqrt[3]{2} + \sqrt[3]{3}$

(D) e

Space for rough work

Matrix Match Types

Each of 2 tables with 3 Columns and 4 Rows has three questions. Column-1 will be with 4 rows designated (I), (II), (III) and (IV). Column-2 will be with 4 rows designated (i), (ii), (iii) and (iv). Column-3 will be with 4 rows designated (P), (Q), (R) and (S). Each question has four options with only **ONE** correct

Answer questions 44, 45 and 46 by appropriately matching the information given in the three columns of the following table.

Let $S = \{A: A = [a_{ij}]_{2\times 2}, a_{ij} \in \{0, \pm 1, \pm 2\}\}$, column-1 and column-2 contain some conditions satisfied by a matrix $A \in S$. Each entry in column-3 is the number of matrices A in S that satisfy one condition from column-I and one condition from column-II

Column–1	Column–2	Column–3
(I) $A = A^T$	(i) $\|A\| = 0$	(P) 5
(II) $A = -A^T$	(ii) $\|A\| \neq 0$	(Q) 8
(III) $A^2 = I$	(iii) $Tr(A) = 0$	(R) 17
(IV) $AA^T = I$	(iv) $Tr(A) \neq 0$	(S) 20

Where A^T, $|A|$ and $Tr(A)$ denote the transpose of A, the determinant of A and the trace of A respectively

44. Which of the following is the only correct combination?
(A) (I) (ii) (R) (B) (II) (iv) (Q)
(C) (III) (iii) (S) (D) (IV) (iii) (P)

45. Which of the following is the only correct combination?
(A) (III) (i) (R) (B) (I) (iii) (S)
(C) (II) (ii) (P) (D) (IV) (ii) (Q)

46. Which of the following is the only correct combination?
(A) (II) (iii) (Q) (B) (IV) (iv) (P)
(C) (III) (iv) (P) (D) (I) (i) (R)

Space for rough work

Answer questions 47, 48 and 49 by appropriately matching the information given in the three columns of the following table.

Let $f(x)=\begin{cases} x+a & x<0 \\ |x-1| & x\geq 0 \end{cases}$, $g(x)=\begin{cases} x+1 & x<0 \\ (x-1)^2+b & x\geq 0 \end{cases}$. Where a and b are non negative real numbers

Column-1 contains information about g(f(x)). Column-2 contains information about interval of g(f(x)). Column-3 contains information about continuity and differentiability of g(f(x)) in R match the following Column(s)

Column–1	Column–2	Column–3
(I) g(f(x)) = x + a + 1	(i) $x \geq 1$	(P) g(f(x)) is continuous for all x then a = 1, b = 1
(II) g(f(x)) = x^2 + b	(ii) $x < -a$	(Q) g(f(x)) is continuous for all x then a = 1, b = 0
(III) g(f(x)) = $(x + a - 1)^2$ + b	(iii) $-a \leq x < 0$	(R) g(f(x)) is differentiable at x = 1 for a = 1, b = 0
(IV) g(f(x)) = $(x - 2)^2$ + b	(iv) $0 \leq x < 1$	(S) g(f(x)) is non-differentiable at x = –1 for a = 1, b = 0

47. Which of the following is the only correct combination?
(A) (I) (ii) (Q) (B) (II) (i) (P)
(C) (III) (iv) (R) (D) (IV) (iii) (S)

48. Which of the following is the only correct combination?
(A) (I) (ii) (P) (B) (II) (ii) (R)
(C) (III) (iii) (Q) (D) (IV) (iv) (S)

49. Which of the following is the only correct combination?
(A) (III) (iii) (R) (B) (II) (ii) (Q)
(C) (III) (ii) (P) (D) (IV) (i) (S)

Space for rough work

SECTION – C

(One Integer Value Correct Type)

This section contains **5 questions**. Each question, when worked out will result in **one integer** from 0 to 9 (both inclusive).

50. Consider a parabola $4y = x^2$ and point B(0, 1). Let $A_1(x_1, y_1)$, $A_2(x_2, y_2)$, ... $A_n(x_n, y_n)$ are n points on the parabola such that $x_r > 0$ and $\angle OBA_r = \frac{r\pi}{2n}$ (r = 1, 2, 3, ..., n) then $\pi\left(\lim\limits_{x\to\infty} \frac{1}{n}\sum\limits_{r=1}^{n} BA_r\right)$ is equal to _____

51. Let N be the number of function f: A $\to$ A where A = {1, 2, 3, 4, 5}, such that f({1, 2, 3}) and f(f({1, 2, 3})) are disjoint. Let M be the sum of all the prime factor of N, then the value of $\sqrt{M}$ is equal to _____

52. The number of solutions of $\log\left(x^{2018}+1\right)+\log\left(1+x+.....+x^{2016}\right)=\log 4034+2017\log x$, is _____

53. If the solution of the differential equation $y\left(2x^4+y\right)\frac{dy}{dx}=\left(1-4xy^2\right)x^2$ is $k(x^2y)^2 + y^3 - x^3 = c$, then k is equal to _____

54. ΔABC is an equilateral triangle with circumcentre (1, 2) and vertex A lying on the line 5x + 2y = 4. A circle with centre I(2, 0) passes through the vertices B and C and intersects the sides AC and AB at D and E respectively. If area of quadrilateral BCDE is Δ, then find the value of $\frac{\Delta}{10\sqrt{3}}$ is _____

Space for rough work